RAJALAKSHMI ENGINEERING COLLEGE
B.E. Computer Science and Engineering · Cyber Security

CR23532

Cyber and Digital Forensics

Case Study Assignments – II | Mini Project

Student name	BARATH RAJ M
Register number	2116241901015
Department	CSE (Cyber Security)
Submitted on	16 October 2026

The five case studies at a glance

Part	Unique scenario title
2(1)	Weekend Copy of an Internal-Assessment Question Bank from a Department NAS
2(2)	Shared Staff-Room PC and a Disguised Printer-Helper Executable
2(3)	Seizure of a Research-Lab Laptop and Portable SSD in a Suspected Dataset Theft
2(4)	Deleted Placement Offer Tracker Recovered from Volume Shadow Copies
2(5)	Tampered Hackathon Registration Through a Student Account

This submission answers all 38 questions across the five case studies, in the same order as the handout. Each case has its own original scenario, a stated set of assumptions, and the deliverables the handout asks for: evidence tables, flowcharts, timelines, a chain-of-custody form, correlation matrices and a sample report outline.

ASSIGNMENT 2(1) / UNIQUE SCENARIO C1-QBANK-14

Weekend Copy of an Internal-Assessment Question Bank from a Department NAS

Case scenario (hypothetical): On Sunday 11 October 2026 at about 7:15 p.m., the audit alert on the CSE department's network storage unit DEPT-NAS-02 shows that the folder IA_QuestionBank was opened and a large archive, qbank_IA2.zip, was read, although the building was closed. On Monday morning a lab assistant finds a 64 GB pen drive, PD-QB1, still plugged into lab workstation LAB-PC-21. The inquiry committee allows the team to examine the NAS logs, LAB-PC-21 and the pen drive.

Working hypotheses and assumptions

Before listing the phases, I would write down what the team is trying to prove or disprove. **H1:** a person used LAB-PC-21 to read qbank_IA2.zip from the NAS and copy it to PD-QB1. **H0:** the activity has an innocent explanation, such as a scheduled backup, a faculty member working remotely or a security scan, and the pen drive is unrelated. The evidence must be able to separate the two.

- Auditing was switched on for the IA_QuestionBank share, and the NAS keeps its logs for at least 30 days.
- Staff and students sign in to lab machines with individual domain accounts; there is no shared lab password.
- The committee's written order covers the three named items plus domain-controller and DHCP logs for 10–12 October only.
- The NAS and workstation clocks are believed to be NTP-synchronised, but this will be checked and not trusted.

Q1. Identification, collection, examination, analysis and presentation

Phase	What the team does in this case	Output
Identification	Log the complaint and fix the time window (11–12 Oct). List the NAS, LAB-PC-21, PD-QB1, domain-controller and DHCP logs, the lab register and CCTV. Ask the administrator to stop log rotation.	Source list, scope note, case number
Collection	Photograph the scene and record the USB port and power state. Image the workstation disk and the pen drive behind write blockers. Export NAS and domain logs read-only. Hash every output; seal and log custody.	Images, log exports, hash manifest, custody forms
Examination	Parse NAS audit entries, Windows event logs, registry USB history, recent-file shortcuts and file-system metadata on the copies. Extract only what is within scope.	Extracted artefacts, each tied to an exhibit ID
Analysis	Merge the artefacts into one timeline. Compare the SHA-256 of qbank_IA2.zip on the NAS with any copy on PD-QB1. Test H1 against H0 and look for contrary evidence.	Timeline, correlation table, assessment of H1 and H0
Presentation	Write the report, keeping facts, inferences and limits apart. Brief the committee in plain language.	Report, appendices, briefing

> ◆ **GOLDEN RULE**
> Every phase ends with a signed hand-over record. The report keeps four claims apart: the file was accessed, the file was copied, the pen drive left with someone, and a named person did it. Proving one does not prove the next.

Q2. Investigation flowchart

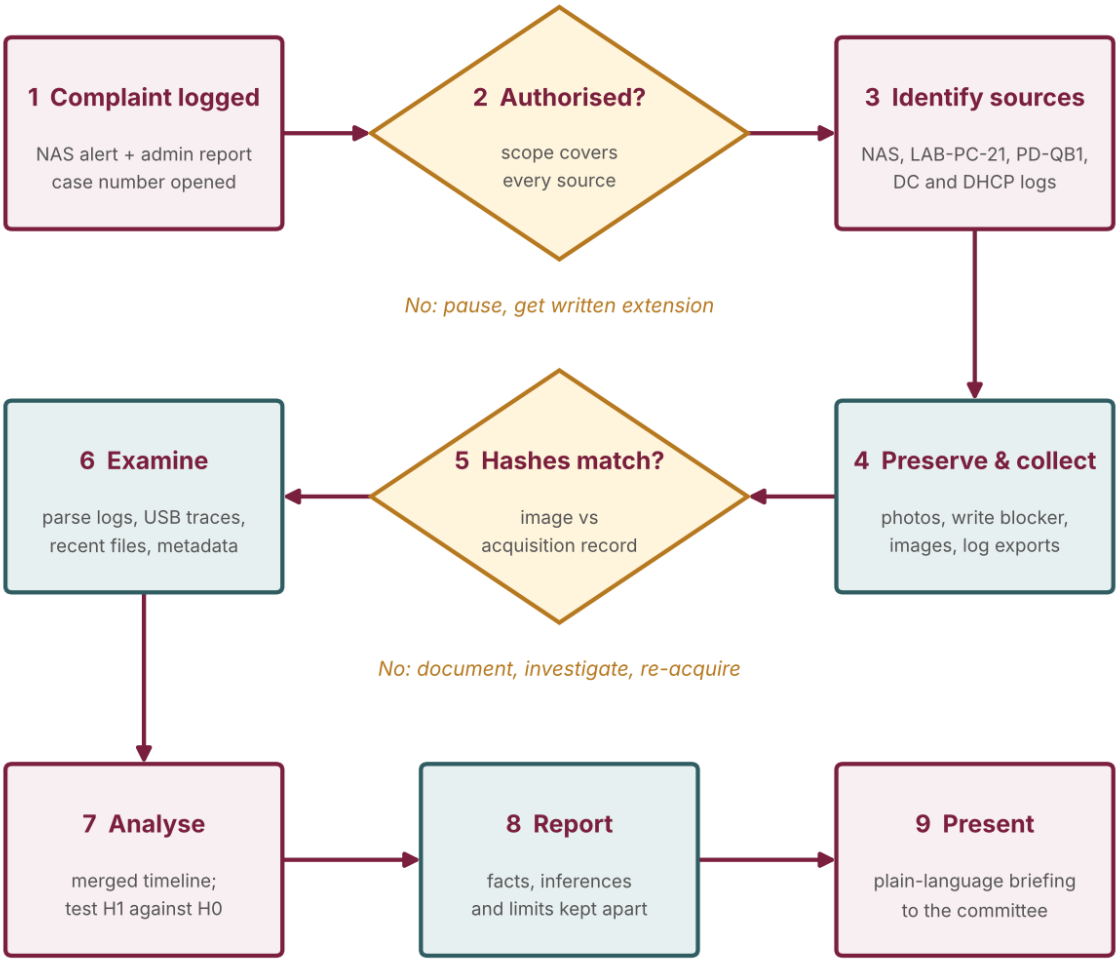


Figure 1. Complaint-to-presentation flow for the question-bank incident, with its two control decisions.

Two decision points keep the process honest. If the written authority does not cover a source, the team pauses and has the order extended instead of looking anyway. If an image hash does not match its acquisition record, the mismatch is documented and investigated before any analysis result is relied on. If a finding ties only to an account and not to a person, the report says so.

Q3. Potential digital evidence and its investigative value

I classify each item as **Primary** (directly records the access or the copying), **Corroborating** (supports or helps attribute the primary evidence) or **Contextual** (explains the background but proves little alone).

ID / item	What it may show	Class	Main caution
E01 Administrator complaint and NAS alert e-mail	When the activity was noticed and which share was involved	Contextual	Discovery time is not incident time
E02 NAS audit log for IA_QuestionBank	Account, source host/IP, open and read events on qbank_IA2.zip	Primary	A read event does not prove a copy was made
E03 LAB-PC-21 disk image	USB connection records, recent-file and jump-list entries, logon sessions, event logs	Primary	Shared lab PC: a profile is not a person
E04 PD-QB1 image	Presence of qbank_IA2.zip or fragments, volume serial, file timestamps	Primary	Other people may also have written to the drive
E05 qbank_IA2.zip on the NAS (metadata, SHA-256)	Reference hash to compare with any copy on the drive	Primary (comparison)	A hash match shows same content, not who copied
E06 Domain-controller and DHCP logs	Which account signed on at which machine and IP lease	Corroborating	Lease reuse, stale or rotated logs
E07 Lab register, CCTV, door-access log	Who was physically present around 18:50–19:30	Contextual	Presence is not keyboard use; clocks differ

What I would preserve first. Anything that rotates or disappears quickly comes first: NAS audit entries, domain-controller logs and DHCP leases, then the state of the workstation. Disk images follow. Even if only a few extracts reach the final report, the complete acquisition stays sealed so that another examiner can repeat the work.

Q4. Collection and preservation precautions

Area	What I would do	Why it matters
Authority and privacy	Confirm the written order. Limit searches to the stated folders, accounts and dates. Leave unrelated student and staff data untouched.	Out-of-scope collection can be challenged and harms privacy
Scene record	Photograph the PC, ports, the pen drive in place, the screen and labels. Note the power state and the time shown on the machine.	Shows the condition at discovery
Contamination control	Never browse the original disk. Use a hardware write blocker, image first and examine copies.	Stops accidental changes to timestamps
Volatile data	If the PC is running, decide and record whether capturing memory or session state before power-off is justified.	Memory can hold session traces that vanish at shutdown
Integrity	Calculate SHA-256 for each image and log export at acquisition and again after transfer. Keep tool logs.	Lets anyone confirm nothing changed
Packaging and storage	Unique exhibit ID, tamper-evident seal, anti-static bag for the pen drive, locked storage with an access register.	Keeps custody unbroken
Honest notes	Record read errors, clock offsets and any action taken on a live system.	Disclosed gaps hurt less than discovered ones

Q5. Building a timeline of the suspected unauthorised access

I would record every event in one worksheet with five fields: original timestamp, source time zone, UTC equivalent, exhibit ID and a one-line meaning. Before merging, I would compare the NAS, the domain controller and LAB-PC-21 against a trusted time reference and note each offset, so that events from different machines are not placed in the wrong order. The entries below are illustrative (IST = UTC+05:30).

Local time (IST)	UTC	Possible event	Source	Shows / does not show
11 Oct 18:52	13:22	Account fac.rk signs on at LAB-PC-21	E06, E03	Account session on the PC; not who typed
11 Oct 19:03	13:33	Pen drive PD-QB1 first connected	E03	Device attached; no proof of transfer
11 Oct 19:11	13:41	NAS session from the PC's IP to IA_QuestionBank	E02, E06	Network path to the share and account used
11 Oct 19:15	13:45	qbank_IA2.zip read in full	E02	Read access; destination unknown
11 Oct 19:18	13:48	Same-size file created on PD-QB1	E04, E05	Strong transfer sign if SHA-256 is identical
11 Oct 19:26	13:56	Account logs off; drive stays in the port	E06, E03	Session ended; drive left behind
12 Oct 09:05	03:35	Lab assistant finds PD-QB1 in the port	E01	Discovery, not incident time

These rows are possibilities, not recovered facts. The sequence becomes convincing only when separate sources agree, for instance when the NAS read at 19:15 sits between the USB connection at 19:03 and a matching file on the drive at 19:18. If the NAS clock were found to run three minutes fast, every NAS entry would be shifted before comparison, and that correction would itself be documented.

Q6. The final forensic report and presenting to a non-technical authority

Section	Content
1 Authorisation and scope	Who ordered the work, the authority relied on, the time range and any exclusions
2 Evidence register	Exhibit IDs, seal numbers, owners, sources, acquisition details, hash values and transfer records
3 Methods and tools	Write blocking, imaging and export steps; tool names and versions; integrity checks
4 Observations	Timestamped events with exact exhibit references, kept separate from opinion
5 Analysis	H1 against H0, the file-hash comparison, account correlation and alternative explanations
6 Conclusions	What is established, probable or not established, with the level of confidence
7 Limitations	Clock offsets, missing logs, shared accounts, anything not examined
8 Appendices	Hash manifest, log extracts, custody forms, approval letter, diagrams

◆ **SAMPLE FINDING F-01**

The sample records would show that an account signed on to LAB-PC-21, a pen drive was connected, and the question-bank archive was read from the NAS between 19:03 and 19:18 on 11 October. If the archive on PD-QB1 carries the same SHA-256 as the NAS original, the copy is established. Linking the activity to a particular individual would still need independent evidence such as the lab register, CCTV or an interview.

Speaking to the committee. I would open with a one-page summary covering what was reported, what the evidence shows, what it does not show and what is recommended. A short timeline carries most of the story. Technical terms get everyday comparisons: a hash is a fingerprint of the file, and chain of custody is the signed visitors' book of everyone who handled the exhibit. A sentence like “the account fac.rk read the archive at 7:15 p.m.” is safer than naming a person as the thief.

Recommended next steps. Restrict the share and rotate affected credentials only after evidence is preserved, ask the account holder to explain the session, and consider keeping question banks offline until the examination is over.

Q7. Legal and ethical considerations in a college environment

The evidence here is sensitive: it includes examination material and personal information about staff and students. The team therefore needs written authority, a limited purpose and careful handling. Permission to examine a college server does not automatically allow reading unrelated private messages. The legal points below should be confirmed with the college's legal adviser: unauthorised access under the Information Technology Act, 2000 (for example Sections 43 and 66); admissibility of electronic records under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023, which expects a certificate describing how the record was produced; and the Digital Personal Data Protection Act, 2023 for personal data.

- **Proportionality:** collect only what the order covers; do not alter original files or circulate raw evidence over personal e-mail or messaging apps.
- **Fair treatment:** the account holder should be able to explain the session, and evidence pointing away from the suspicion is preserved with the same care.
- **Confidentiality:** the question bank itself is confidential, so access to exhibits is limited to named people and personal details are redacted in committee copies.
- **Independence:** an investigator with a conflict of interest, such as a colleague of the person concerned, should not lead the examination.
- **Retention:** record who may open the evidence, how long it is kept and how verified copies are released to an authorised body.

Claim	What would support it	What would remain uncertain
The file was accessed	NAS audit event for a named account on qbank_IA2.zip	Credential misuse or a service account
The file was copied to the pen drive	Identical SHA-256 on PD-QB1 plus a USB connection on the same PC in the same window	Who plugged the drive in
A named person did it	Lab register or CCTV placing the person at LAB-PC-21, consistent login pattern, and the person's own explanation	Shared login, shoulder-surfing, spoofing

At this stage the matter is an allegation. Following this process lets the college reach a careful decision without exposing the question bank or blaming a person before the evidence has been checked.

ASSIGNMENT 2(2) / UNIQUE SCENARIO C2-STAFFPC-22

Shared Staff-Room PC and a Disguised Printer-Helper Executable

Case scenario (hypothetical): FAC-PC-05, a desktop in the CSE staff room, is used by five faculty members, a part-time technician account (tech.guest) and a print-management service. During a routine check IT staff find prnhelper_x64.exe inside the hidden folder C:\ProgramData\.spool_cache, together with a password-protected spreadsheet, internal_marks_bk.xlsx, and browser profiles under several accounts. The inquiry wants to know which account created or used these items, using ILookIX for the analysis.

Q8. Hardware and software documented before analysis

Item	What the examiner records	Why it matters
Computer	Make, model, asset tag, serial number, BIOS/UEFI version and mode, photographs, attached peripherals	Provenance and boot behaviour
Storage	SSD or HDD model and serial, capacity, interface, partition layout, sector size, any encryption	Selects the correct imaging method and states limits
Memory and power	RAM size, power state, boot method, external media connected	Volatile artefacts that may exist
Operating system	Windows edition and build, install date, locale, time zone, clock offset	Parsing rules and timestamp context
File system	NTFS volumes, ACLs, $MFT and USN journal availability, volume serials	Metadata that can be recovered
Accounts	Local and domain accounts, SIDs, profile paths, last-logon data	Needed to tie artefacts to an account
Software and security tools	Installed programs, AV/EDR, disk-encryption tool, browsers, scheduled tasks, services	Benign or suspicious execution clues
Network	Host name, IP/MAC, domain membership, proxy or VPN settings	Cross-system attribution

Most of this can be taken from scene notes and a read-only image. I would not boot the original machine just to fill in an inventory field.

Q9. How storage, operating systems, file systems and metadata help locate evidence

Layer	What it offers	Use in this case
Storage device	A sector-level copy, including deleted entries and unallocated space; model and serial identify the exact source	Unallocated space may hold remnants of earlier versions of prnhelper_x64.exe; SSD TRIM can limit this
Operating system	User profiles and SIDs, event logs, registry hives, Prefetch, scheduled tasks, installed programs	Run keys, a scheduled task and execution traces for the executable
File system (NTFS)	$MFT records, creation/modify/access/record-change times, alternate data streams, USN journal, owner and ACLs	When and where the executable appeared, and which SID owns it
Metadata	Document properties, PE headers, hashes, file signatures	Author field of the spreadsheet; compile time and signature status of the executable

No single layer proves much. A timestamp alone cannot show that anyone ran the program. For prnhelper_x64.exe I would compare the NTFS record with browser download history, recent-file shortcuts, execution traces and scheduled tasks, and check whether the file was renamed or moved.

Q10. Hidden and encrypted files in an investigation

A hidden file is not necessarily malicious. The hidden or system attribute, a dot-prefixed folder name, an alternate data stream or a renamed extension can all keep a file out of ordinary folder views, and print software often does this legitimately. I would examine hidden and system folders, alternate data streams and deleted entries on the image, leaving the originals unchanged. A confusing name is a reason to look closer, not proof of concealment.

Encryption is a different problem: the file exists but its contents cannot be read. It might be a password-protected document, an encrypted archive or a protected volume. I would not state what internal_marks_bk.xlsx contains until it has been lawfully opened. Even without decryption, the name, size, timestamps, recent-file links and application traces can help answer the case questions.

- Record each relevant file's full path, volume and the image it came from.
- Give extracted files their own SHA-256 values and log any step that changes the form, such as decryption or decompression.
- If a password or key is provided with proper authority (for example by the owner, or recovered from a managed key store), use it on a duplicate and keep both the encrypted and decrypted versions under separate IDs.
- Do not attempt password guessing without approval, and store any keys securely with a record of who accessed them.

Observation / test	Interpretation if confirmed	Competing explanation
The .spool_cache folder has hidden and system attributes and was created during a tech.guest session	Possible concealment or staging	Print software creates hidden cache folders
prnhelper_x64.exe has no valid digital signature and its hash is not on the vendor list	A reason for deeper analysis	An old or in-house utility
internal_marks_bk.xlsx is encrypted and its recent-file link appears only under tech.guest	Interaction by that account	A backup or sync job touched the file
A high-entropy file without extension sits in a temp folder	Possible container or archive	Compressed print spool data

Before putting a hidden or encrypted file into the report, I would ask whether it actually helps answer the allegation. Unrelated personal material stays preserved in the image but is not disclosed in the report.

Q11. Evidence-linkage table: linking files to a user

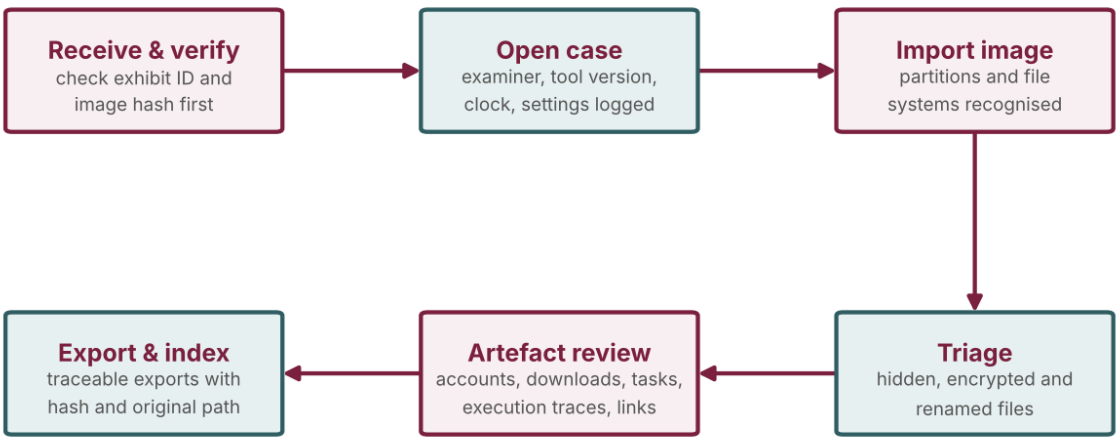
The entries below are illustrative. They show how I would test a possible link between the executable, the spreadsheet and one Windows account.

Source artefact	Illustrative finding	Links to	Strength / caveat
NTFS $MFT / USN journal	prnhelper_x64.exe created 12 Oct ~14:20, owner SID of tech.guest	Account	Medium–high: ownership can be inherited from the creating process
File name and path	C:\ProgramData\.spool_cache\	Location, concealment	Low: a path alone says little
Windows logon events	tech.guest interactive session ~14:05–14:50	Account	High for the account, not for the operator
Browser downloads	tech.guest Edge profile records the download of the same file name at ~14:17, matching hash	Origin of the file	Medium: a profile can be used by others
Prefetch / Amcache	Execution at ~14:22, run count 3	Execution	Medium: depends on OS settings
Scheduled task	Task “SpoolSync” starts the executable at logon; author field names tech.guest	Persistence	Medium: review creator and trigger
Recent files / jump lists	Link to internal_marks_bk.xlsx under tech.guest	Interaction with target	Medium: shows an open attempt, even on an encrypted file
Door or attendance log	tech.guest badge used at ~13:58	Physical presence	Supporting only; others can share a badge

When matching records I would prefer stable identifiers: hashes over file names, SIDs over display names, and timestamps converted to one time zone. One plausible sequence is a download at 2:17 p.m., file creation at 2:20 p.m., execution at 2:22 p.m. and an attempt to open the spreadsheet afterwards. I would only connect these events if the matching artefacts exist and agree.

Q12. Importing and analysing the evidence with ILookIX

ILookIX belongs to the ILook family of forensic tools, and menus and supported image formats vary between releases. I therefore describe the checks an examiner must make instead of inventing screenshots. If ILookIX were not available, a tested alternative could be used, and the report would name it and explain the substitution.



Importing a verified image is not the same as acquiring the original disk.

Figure 2. Intended ILookIX workflow from verified image to traceable exports.

1. Receive the authorised evidence, check exhibit IDs and verify the hash of the disk image. Importing an image is not the same as acquiring the original disk.
2. Create a new case and record the examiner, the ILookIX version, the computer clock and the settings used.
3. Import the verified image, note the partitions and file systems the tool recognises and record any error messages. If another image format is needed, keep both the original and the converted copy.
4. Triage: list hidden and deleted files, encryption indicators, and files whose content does not match their extension. Hash prnhelper_x64.exe and compare it with a trusted copy if one exists.
5. Review user profiles and SIDs, browser downloads, shortcuts, execution traces and scheduled tasks, recording the search terms and dates used so the search can be repeated.
6. Export relevant artefacts with their original paths, timestamps, source-image details and hashes. Treat every export as a separate evidence item.
7. Compare the results with independent logs and account records, then prepare an evidence index another examiner can check.

◆ **EXPECTED OUTPUTS**
An image-import record; a partition and account inventory; a filtered file listing; the executable's hash and metadata; account-to-artefact links; an export manifest; and an analysis log. These are intended outputs, not observed results of a tool run.

Validation. Before drawing conclusions, I would reopen a sample of exported files in a second viewer and re-check their hashes. I would also list anything the tool could not parse, such as encrypted content or unsupported log formats, rather than treating missing results as proof that nothing happened.

Q13. Direct evidence versus supporting or contextual evidence

A direct artefact records the activity under investigation and names an account. A reliable process-execution record with an SID, for example, links the program run to that account. Even so, it does not show which human was at the keyboard. A file in someone's profile, a duty roster or a matching IP address is usually only supporting evidence because each has several explanations.

Example claim	Direct, account-linked artefact	Supporting or contextual information
An account ran the executable	Trusted process-execution audit keyed to the account SID and executable hash	Browser download in the same profile; Prefetch entry on its own
The executable touched the spreadsheet	File-open audit tying the process to the object	Timestamps and a recent-documents shortcut
A named person operated the PC	Independent authentication (for example a personal token or biometric) plus corroboration	Duty roster or badge presence alone

Q14. Limitations and alternative explanations

- **Shared access:** a colleague could have used an unlocked session, a saved browser login or a remote-control tool, even though the account belonged to someone else.
- **Unreliable time order:** copied files, restored backups, delayed writes or a wrong clock can change how events appear on a timeline.
- **Automated activity:** updates, antivirus scans, scheduled tasks and backup agents can create files or launch programs without a user clicking anything.

- **Missing evidence:** cleared browser history, overwritten logs, disabled auditing and SSD TRIM may hide useful context.
- **Tool limits:** ILookIX or any other tool may not parse every artefact, so a negative result is not proof of absence.

Balanced conclusion for C2-STAFFPC-22. If browser, NTFS and execution evidence all point to one Windows account, I could reasonably report an account-level link. I would not name the account owner as the operator until shared access, automation and remote sessions have been considered. The report should identify the executable by its hash and state clearly whether its behaviour was actually tested.

ASSIGNMENT 2(3) / UNIQUE SCENARIO C3-RLAB-LAP-09

Seizure of a Research-Lab Laptop and Portable SSD in a Suspected Dataset Theft

Case scenario (hypothetical): On Tuesday 13 October 2026 at about 3:30 p.m., a laptop, RL-LAP-09 (internal NVMe SSD), and a 500 GB portable SSD, PSSD-03, are found on a research-lab desk after a complaint that an unpublished dataset was being copied for an outside organisation. The laptop lid is closed and a small LED blinks slowly, so it may be asleep rather than off. The priority is to collect and image both devices without altering them.

Q15. Collecting and transporting the laptop and USB device

1. Confirm the written authority, the case number and the exact location it covers. Record who is carrying out the seizure.
2. Photograph the desk, both devices, cables, charger, ports and any labels before touching anything. Record the LED behaviour and who had access to the room.
3. Do not open the lid, press keys or connect the laptop to a network. Treat a blinking LED as a sleep state, not as “off”.
4. Decide the power-state handling with a trained examiner. If disk encryption is likely and live capture is authorised, record why memory or session state is being captured first; otherwise follow the documented shutdown procedure. Keep the charger connected so the battery does not drain mid-decision.
5. Note the connection of PSSD-03 (port, cable) and remove it only after that decision. Package it as its own exhibit.
6. Record make, model and serial numbers for both items. Apply tamper-evident seals and give each device a separate exhibit ID.
7. If the laptop stays powered and online, isolate it from the network by a documented method or a shielded bag so that no remote-wipe command can arrive.
8. Transport in padded, anti-static packaging, upright, away from heat, magnets and vibration, with two people present. Enter each hand-over in the custody form and place the sealed exhibits in secure intake at the lab.

◆ **SLEEPING-LAPTOP CAUTION**
A dark or silent screen does not prove the laptop is off. Sleep or hibernation, hardware-backed encryption and modern SSD behaviour can change the correct acquisition decision. Record the choice made and why the risk was accepted.

Exhibit identifiers. RL-01 for the laptop and RL-02 for the portable SSD; their images would be RL-03 and RL-04. A working copy gets a linked ID such as RL-03-W1, so every later result can be traced to its source.

Q16. Physical acquisition and why a forensic image is used

Physical acquisition copies the addressable sectors of a storage device, not just the files visible in File Explorer. It captures active files, file-system structures and remnants in unallocated space. I would normally examine the verified image rather than the original because ordinary software can write temporary files,

update access times or rebuild indexes and thumbnails, and any of these would change the evidence. A hashed image also lets a second examiner repeat the work.

A physical image does not guarantee that every old byte is available. SSD TRIM, encryption, unreadable sectors and firmware-managed areas may prevent full recovery. I would state precisely whether the acquisition covered the whole disk, one partition or only selected files.

Q17. IXImager-based imaging workflow

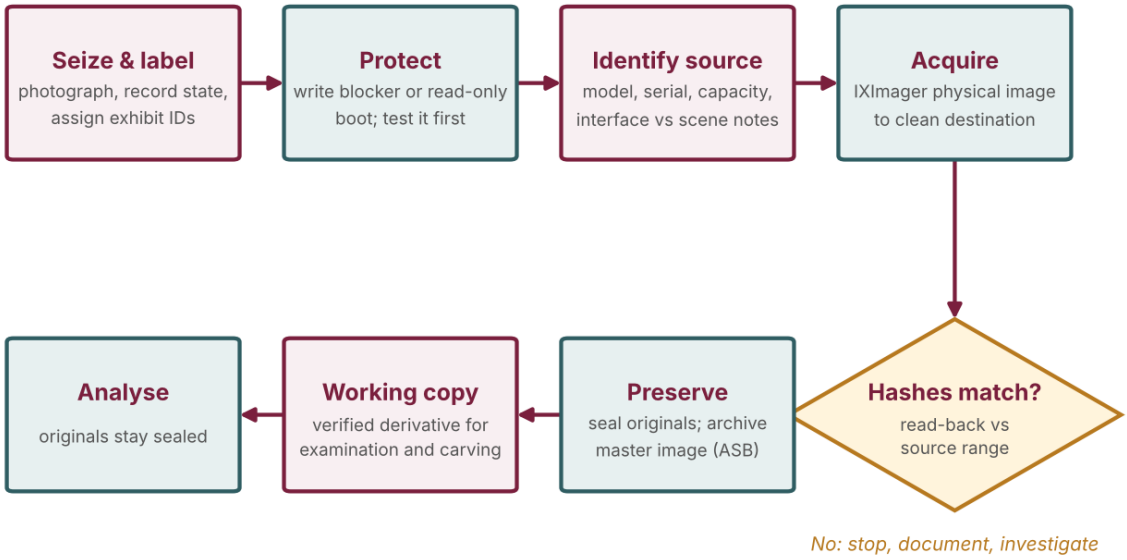


Figure 3. Imaging stages and the exception branch for read errors or hash mismatches.

IXImager is the imaging tool associated with the ILook family. I describe its workflow from its documented purpose and do not claim that it was run. In a real laboratory I would first confirm which version is available and whether it supports the evidence devices and the intended image format. For the internal NVMe drive, a suitable write-protected adapter or a write-protected forensic boot environment would be needed, or the drive would be removed and connected through a blocker.

Gate	Procedure	What is recorded
Source identification	Check model, serial, interface, capacity and partition signs; compare against scene notes	Source address, device inventory
Write protection	Use a tested hardware blocker or documented read-only equivalent; prevent read-write mounting of the source	Blocker model, test result
Destination	Prepare clean, adequately sized storage with a separate case folder and access control	Destination ID and capacity
Imaging	Select the physical source and a supported image format; run the acquisition	Tool build, operator, start/end time, errors, segment files
Verification	Run the tool's verify function and an independent read-back; compare sector coverage and hashes	Hash values, verification result, logs
Preservation	Seal the originals, archive the master image, make a derivative working copy	Manifest and transfer entries

The laptop and the portable SSD are imaged separately, each with its own serial number, log, hash and custody history.

Q18. How hash values show integrity before and after acquisition

A hash function gives the same SHA-256 value for the same bytes and a completely different one if even one bit changes. I would calculate and record hashes at defined points: at acquisition, after verification, after copying to archive, and before and after analysis. The comparison must be between the same kind of data. The hash of a compressed evidence container, for example, will not equal the hash of the raw disk data inside it, and both can be correct if their meaning is documented.

Checkpoint	Value recorded	Decision rule
At acquisition	Source-read hash if the tool provides it, with sector range and errors	Record the baseline and its limits
After image creation	Hash of the reconstructed raw data or tool verification result	Compare like with like
Master archive	SHA-256 of each image container and segment	Manifest of unchanged master files
Before and after analysis	SHA-256 of the read-only image and of the working copy	Investigate any unexplained change

◆ **IF READS FAIL**
If some sectors cannot be read, no examiner should report a perfect source-to-image match. Document the unreadable block addresses, the retries and the filler policy, and state that verification applies only to the content successfully captured.

Q19. Role of the ASB container

The syllabus presents the ASB container as the way acquired data is kept together with authentication information so that compatible software can open it and detect changes. I would treat it as a managed evidence package rather than a plain file. The forensics wiki lists IDIF, IRBF and IEIF as IXimager's native formats and does not describe ASB, so before relying on ASB I would confirm that the version of the tool in use supports it. The extension alone cannot guarantee integrity or long-term compatibility.

Category	Information that should accompany the container
Identity	Case number, exhibit ID, device make/model/serial, and image ID
Acquisition	Examiner, date and time with time zone, tool name and version, source range, settings
Integrity	Hash values, verification results, segment list, error logs
Custody and storage	Custody entries, storage location, access records, backup copies
Conversions	Any converted copy (for example to raw) and a note explaining why it was made

Q20. Data carving and when it is needed

Data carving searches the raw image for recognisable file structures, such as headers and footers, without relying on the file system. It is useful when directory entries are missing or damaged. It can recover part or all of a deleted document, picture or archive, but it usually cannot restore the original name, folder or owner.

Situation	Is carving likely to be needed?
Deleted file with its $MFT record intact	No: recover through the file system, which keeps name and path
Quick-formatted or repartitioned device	Yes: directory structures are gone but data areas may remain
Corrupt or overwritten $MFT	Yes, for files whose entries cannot be read
Fragmented large files	Partly: simple carving may return incomplete files

Situation	Is carving likely to be needed?
SSD after TRIM	Often little remains, so results may be empty
Encrypted volume without a key	No useful result: carving would only find ciphertext

I would carve only on a verified working copy and save the signatures, offsets and settings used. Each recovered file should be checked for completeness and to see whether it opens correctly. Fragmentation, overwriting and encryption may leave only partial content or nothing at all.

Q21. Date and time problems in digital evidence

Problem	Why it matters	How I would document it
Clock offset or drift	Device time may differ from real time	Photograph or record BIOS/OS time next to a trusted reference; state the offset and uncertainty
IST versus UTC	Different sources store local time or UTC	Keep the raw timestamp and the source zone; convert consistently (IST = UTC+05:30)
File-system semantics	Created, modified, accessed and record-changed times behave differently; copying alters some of them	Name each NTFS field and explain what it means
Precision and delays	Caching, rounding and delayed writes blur exact order	Describe resolution and any gaps
Deliberate manipulation	Someone may have changed the clock	Look for time-service events and independent records; never “fix” the source records

Q22. Chain-of-custody record: RL-01 laptop

This sample form shows how the laptop moves from the collector to the laboratory examiner. The names, times and seal numbers are examples and would be replaced by real entries and signatures in an actual seizure. The portable SSD (RL-02) would have its own form.

No.	Date / time (IST)	Released by → received by	Place and purpose	Seal and condition
1	13 Oct 2026, 15:30	Lab custodian → Collector A	Research lab; initial seizure	Seal RL-S-201 applied; both sign
2	13 Oct 2026, 15:50	Collector A → Transport officer B	Move to forensic lab	RL-S-201 intact; both sign
3	13 Oct 2026, 16:25	Transport officer B → Evidence custodian C	Lab intake and locker	RL-S-201 intact; entry logged
4	14 Oct 2026, 10:00	Custodian C → Examiner D	Authorised imaging session	Seal opened and noted; both sign
5	14 Oct 2026, 13:40	Examiner D → Custodian C	Return to locker after imaging	New seal RL-S-202; both sign

Mandatory fields for every real entry. Exhibit ID, make and serial number, person handing over, person receiving, reason for transfer, date and time with time zone, location, seal condition and signatures. If a seal is opened and replaced, both actions appear in the record.

Acquisition integrity statement. “The laptop and portable SSD were documented and imaged separately through write-protected procedures. The acquisition logs record the exact source range, errors and verification results. The originals are sealed, and examination takes place on working copies.” This wording is deliberate because no acquisition was actually performed for this written case study. If the disk is

encrypted and no authorised key is available, the report should say that part of the evidence could not be examined.

ASSIGNMENT 2(4) / UNIQUE SCENARIO C4-OFFER-VSS-12

Deleted Placement Offer Tracker Recovered from Volume Shadow Copies

Case scenario (hypothetical): A placement-cell workstation, PC-TP-02, held a workbook named OfferTracker_2026.xlsx listing company offers and student package details. On 14 October 2026 the file is found missing after a late-night session by a temporary staff account, tmp.staff. A verified forensic image of the Windows 11 system (NTFS) exists and contains Volume Shadow Copies. The team must decide whether an earlier version can be recovered and what the traces really prove.

Q23. Selecting and categorising evidence before detailed analysis

I would start with focused questions: which version of the workbook existed before the incident, what changed, when, and was the file deleted or only moved? I would preserve the full image but search only the relevant folders, filenames, dates and activity records, so unrelated student data is not examined.

1. Confirm the image ID and hash, the volumes present, the system clock and whether shadow copies are available.
2. Identify likely workbook names and locations, Office recent-file and AutoRecover entries, $MFT and USN records, Recycle Bin entries and related application activity.
3. Group the potential evidence by role: original content, system activity, recovered copies and supporting records. “Suspicious” only means “needs closer checking”.
4. Use hashes to spot duplicate files and exclude ordinary system data, while keeping a record of what was filtered out.
5. Record the exact dates, keywords and filters used, because several shadow copies may hold versions of the same workbook.

Q24. Evidence classification scheme

Class	Examples in this case	Treatment and investigative role
User-created files	OfferTracker_2026.xlsx, draft offer letters, notes	Primary business content; compare approved versions
System and application files	NTFS metadata, Office AutoRecover, Windows event logs, jump lists	Show timing and software behaviour; check parser output
Potentially suspicious files	A renamed copy of the tracker, a cleanup script, an unfamiliar tool	Review behaviour and legitimate uses before labelling
Deleted or recovered artefacts	Recycle Bin entries, shadow-copy versions, unallocated fragments	Recover read-only; record origin of each item
Supporting and contextual evidence	Cloud-sync logs, browser history, user sessions, badge records	Corroborate transfers or account timing

Q25. How Event Analysis helps reconstruct activities

Event analysis turns scattered records into one ordered sequence. In this case the USN journal may show file changes, Office records may show a recent edit, Windows security logs may show which account was

signed in, and sync logs may show another copy. I would combine them in a timeline that keeps the raw time and its UTC equivalent, and also note what each artefact can actually prove. A “last modified” time does not describe every edit, a browser visit does not prove an upload succeeded, and a deleted entry can result from an ordinary save operation.

Source	Event type	What it adds to the sequence
$MFT / $UsnJrnl	Create, rename, delete, data change	Order of file-system events and the reason codes
Office recent files / AutoRecover	Open and save	When the workbook was last handled in the application
Windows Security events	Logon, logoff, object access	Which account was active in the window
Recycle Bin ($I / $R)	Deletion by normal route	Original path and deletion time, or its absence
Sync or cloud client logs	Upload, share, version	Whether another copy exists elsewhere

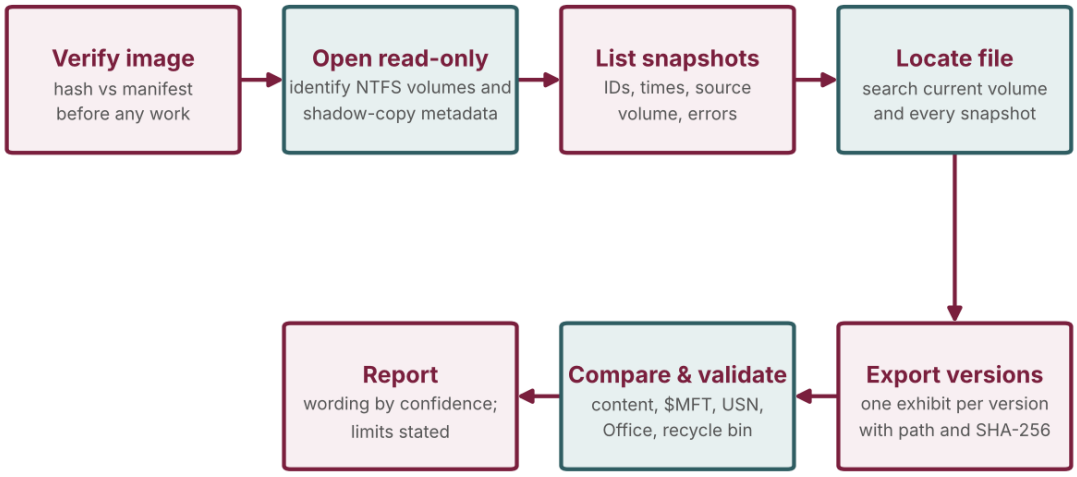
Q26. Cloud Analysis and Lead Analysis

Method	Questions it helps answer	Representative records	Caution
Cloud Analysis	Was the workbook synchronised, shared, versioned or uploaded? Is there a retained copy off the PC?	Authorised provider audit logs, revision history, object IDs, access logs	Provider timestamps and retention differ from the local PC
Lead Analysis	Which clue deserves the next validated search? Who or what else is connected to this item?	File-name variants, user names, USB IDs, recipient domains, unique codes	A lead is only a hypothesis until corroborated
Together	Could a local deletion coexist with a cloud-retained copy?	Shadow-copy version, sync time, cloud revision	Same name or size does not prove same content

Example lead workflow. Suppose the recovered version contains a unique offer reference, OFR-2026-0417. I would search the approved image for that reference, check related browser or transfer traces, and then request a narrowly scoped cloud audit export if needed. Recording a negative search result is also useful, and it respects privacy better than opening every placement document.

Confidence vocabulary. I would use “observed” for a verified artefact, “corroborated” when two independent sources agree, “plausible” when an explanation has support but alternatives remain, and “unresolved” when evidence is insufficient. Numerical confidence scores would not be appropriate without a defined method.

Q27. Volume Shadow Copy analysis and recovery workflow



A shadow copy is a point-in-time view; it does not show who deleted a file.

Figure 4. Read-only shadow-copy recovery and validation process.

The Volume Shadow Copy Service can keep an earlier state of a Windows volume. If a workbook changed or vanished after a snapshot was taken, that snapshot may still contain the earlier version. A snapshot is not a recording of everything that happened between two times, and older snapshots may already have been deleted.

1. Open the verified image read-only and identify the NTFS volumes and any stored shadow-copy metadata.
2. List the snapshots and record their identifiers, source volumes, capture times and any errors the tool reports.
3. Search for the workbook and related browser or Office records in the current volume and in every relevant snapshot.
4. Export each useful version as its own exhibit with its original path, snapshot ID, metadata and SHA-256 hash.
5. Compare versions using hashes and a careful content review. A different hash means different bytes; the report must still explain the actual changes.
6. Cross-check against $MFT, USN, Office, Recycle Bin and approved cloud records to test whether editing, moving or deleting took place.
7. Save the extraction log, open exports in a second viewer and state any limits caused by missing or damaged snapshots.

◆ **CRUCIAL LIMITATION**
A file present in a snapshot but absent from the current volume shows that the two points in time differ. Without more evidence it does not show the exact deletion time, the deleting process or the person responsible.

Q28. Hypothetical life-cycle timeline of a file

The timeline below is a made-up example in IST. In a real case I would also save the UTC equivalent and note any clock uncertainty.

Time (IST) / artefact	Illustrative state	How it would be checked
08 Oct 10:05 / NTFS	OfferTracker_2026.xlsx created in D:\Placement\	$MFT record, path, workbook properties
10 Oct 16:30 / Office + NTFS	Workbook modified: package details updated	Office revision data and modify time
11 Oct 17:00 / VSS-1	Snapshot captures the workbook (61 rows)	Snapshot ID, export hash
11 Oct 18:10 / sync log	Candidate copy placed in a shared team folder	Provider action and object ID
13 Oct 22:40 / Office + Security log	tmp.staff session saves the workbook (58 rows)	Account SID, logon session, content comparison
13 Oct 23:55 / $UsnJrnl	Delete-type event; no Recycle Bin entry	USN reason code and file reference; possible Shift+Delete or other tool
14 Oct 09:10 / report	Administrator notices the file is missing	Statement; discovery is not incident time
15 Oct 11:20 / forensics	Version from VSS-1 exported read-only	Export log, chain of custody, hash

I would take care not to read more into a USN journal entry than it shows. An Office application can create a temporary file and rename it during an ordinary save, and a file may have been moved instead of deleted. The timeline should be tested against those possibilities.

Q29. Validation of a recovered artefact before it becomes a finding

Check	Pass condition
Source image	Verified against its preserved manifest before and after extraction
Snapshot details	Snapshot ID, volume, capture time and tool build recorded
Location	Recovered path, offset or file reference and object metadata noted
Exhibit control	Exported item has a unique exhibit ID and SHA-256 digest
Format	File signature checked independently of the extension; the workbook opens without silent repair
Content comparison	Compared against any current copy and trusted approved version; similarity stated precisely
Timestamps	Raw values retained and normalised separately; creation inside a snapshot not confused with snapshot time
Disclosure	Parser warnings, corruption, partial fragments and missing proof disclosed

Q30. Sources of error and alternative explanations

- **Snapshot timing:** a snapshot shows the state only at capture time; the workbook may have been edited long before.
- **Normal Office behaviour:** temporary files and rename-on-save operations can resemble deliberate deletion.
- **Timestamp differences:** wrong clocks, time-zone conversion, imported files or retained metadata can put events in the wrong order.
- **Account use:** a signed-in account could have been shared, used remotely or driven by a scheduled task.

- **Missing copies:** old snapshots can be removed by cleanup or storage limits, and TRIM or overwriting can erase recoverable data.
- **Uncertain recovery:** a carved document may lose its name and folder and could be mistaken for a similar file.

Case synthesis. If the shadow-copy records and the exported file are validated, I could report that a particular version of the tracker existed when that snapshot was taken. A later missing file and supporting NTFS records might suggest deletion, but neither fact alone shows dishonest intent. Identifying the person responsible would need separate login, process or other supporting records.

Finding category	Reporting language
Confirmed if validated	"An earlier version of OfferTracker_2026.xlsx was recovered from snapshot VSS-1."
Corroborated if supported	"A file-system delete event and a second source agree on later removal."
Not established	"The current material does not identify who deliberately removed the file."

Closing evidence package. Classification table, snapshot register, original and recovered file details, version comparison, timeline, validation checklist and hashes, limited to the placement data needed to explain the findings.

ASSIGNMENT 2(5) / UNIQUE SCENARIO C5-HACK-REG-17

Tampered Hackathon Registration Through a Student Account

Case scenario (hypothetical): At 10:48 p.m. on Monday 12 October 2026, the registration record of team NullPointers on the college hackathon portal is altered: the team lead is replaced and a new project abstract is uploaded, using the account of a second-year student. The student says they did not do it. With written permission, investigators can examine the student's Windows laptop (LAP-S12), browser profile, e-mail, chat messages, smartphone and the portal's cloud audit records. The question is whether the student made the change or someone else used the account.

Q31. Evidence from Windows and other operating systems

System	Potential evidence	Use in this case
Windows (LAP-S12)	Security events for logon, unlock and logoff; registry; Prefetch and Amcache; recent files and jump lists; $MFT and USN; Wi-Fi profiles; time settings	Shows whether the laptop was in use and by which account before 22:48
Linux, if used	auth.log or journal entries, shell history, browser profile under the home directory	Equivalent session and command evidence
macOS, if used	Unified logs, file-system events, browser profile, keychain metadata	Equivalent session and application evidence
Android / iOS	Covered in Q34	Authentication prompts and app activity

It is important not to treat all logins as the same thing. Being signed in to Windows, having a browser profile open and holding a valid portal session are three separate observations. The report must say which layer the evidence identifies before any conclusion about the student is drawn.

Q32. What browser artefacts reveal and why they need careful interpretation

Browser history can show that the portal was visited, while the cache, downloads, saved logins, form data and extensions give further context. A visited address does not prove the server accepted a submission. Browsers can preload pages, sync history across devices, keep several profiles, run in private mode and reuse a stolen session cookie, so each browser clue has to be checked against the portal's own logs.

Browser artefact	Potential value	Main caveat
History and visit times	Opportunity to reach the portal pages	Preloaded or synced visits can appear
Cache and form remnants	Candidate text of the abstract	May be stale or incomplete
Download list	File obtained before the alleged upload	A download is not an upload
Cookies and session tokens	Authentication context for the portal	Possessing a token does not identify the person
Saved passwords and autofill	How the login could be repeated	Anyone with profile access could use them

Q33. E-mail and messaging evidence and its correlation

E-mail and chat can contain sign-in warnings, password-reset messages, one-time codes, attachments and conversations about the abstract. Message headers, Message-IDs and provider delivery records are generally more reliable for authentication than a screenshot showing only a display name. I would collect only the relevant conversations and preserve their original metadata.

- **E-mail:** headers and Received chain, Message-ID, DKIM/SPF results, sign-in alerts, reset or code messages, attachments.
- **Messaging:** message text, attachments, sent/delivered/read times, group membership, deleted-message traces and cloud backups.
- **Correlation examples:** a one-time-code e-mail at 22:42 against a portal login at 22:43; an abstract file shared in chat compared by hash with the uploaded document; a chat message saying “updated” against the portal change time.

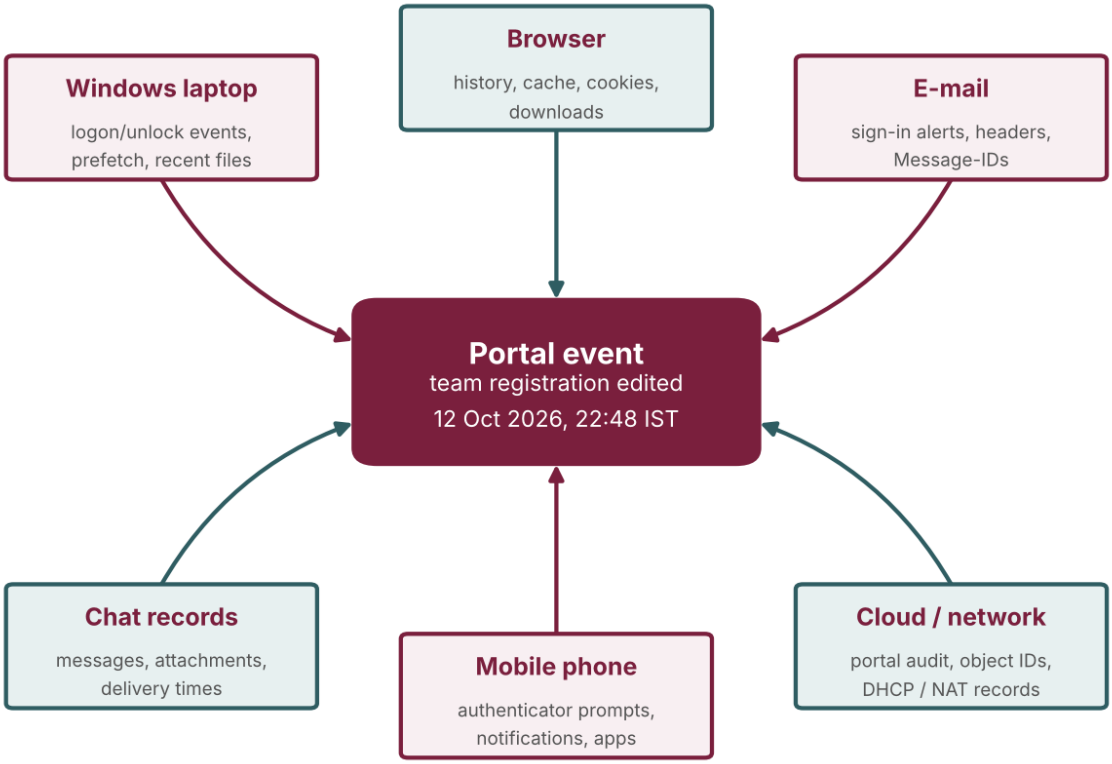
Sending, receiving and reading a message are different events, so I would not assume that one proves another.

Q34. Mobile phone forensic considerations and challenges

- **Initial condition:** photograph the screen, battery and connection status. Any decision to isolate the phone from the network should follow a documented, device-specific plan.
- **Permission and privacy:** establish whether the phone belongs to the student or the college and collect only what the consent or lawful authority covers.
- **Technical limits:** device encryption, locked apps and the absence of a passcode may limit the team to logical data or block collection altogether.
- **Useful material:** where accessible, account identifiers, authenticator or portal-app activity, notifications and downloaded files.
- **Interpretation:** notifications can be delayed or synchronised from another device, and an approved prompt does not by itself show the owner knowingly authorised the action.
- **Handling:** record the extraction method, software version, scope, device identifiers and integrity checks, and avoid browsing unrelated personal apps.
- **Volatility:** remote wipe, automatic updates and message auto-deletion can change data while the phone is connected.

Verification question. If the portal shows a multi-factor approval at 22:44, I would ask which device received it and what exactly was approved, and also whether the laptop already held a valid session. A successful approval can support an account-level event but does not by itself identify the person involved.

Q35. Internet and cloud-based evidence sources



Account use, device use and the identity of the human operator are three separate propositions.

Figure 5. Multi-source evidence map for the portal event; each source is tested separately before synthesis.

The portal provider may hold login results, source IP addresses, user-agent strings, upload times, file hashes, object versions and session-revocation records. Supporting network records might include campus DHCP, VPN or proxy logs. I would request these through proper authority and check the provider's retention limits.

Server records are especially helpful because they show what the portal accepted, not just what a user's device displayed. An IP address is still not a person: shared Wi-Fi, VPNs, NAT and mobile networks make address-based attribution unreliable. Every provider export should state its scope, export time, time zone and responsible administrator.

Preserve first	Scope	Reason
Portal authentication and object audit	10–13 Oct, account and object IDs	Rotating records can be lost
Password-reset and MFA events	Target account, same window	Tests whether the account was compromised
Endpoint and browser state	Authorised profiles and dates	May clarify which device session was used
Cloud object versions	The submitted abstract and its revisions	Authoritative result of the upload
Network allocation records	IP-to-time mapping with precise zones	Contextual source correlation

The aim is to obtain original, traceable exports rather than copying information informally from a dashboard. I would hash exported files, preserve how they were produced and include only the records needed for the incident.

Q36. Cross-source correlation table

This matrix is a worked example of how evidence from the computer, browser, e-mail and chat, mobile and cloud sources might support or challenge each other. The events are not taken from real accounts or devices. Times are IST on 12 October; actual analysis would also record UTC and clock differences.

Source and illustrative time	Hypothetical artefact	Correlation key and meaning	Limitation
Windows laptop / 22:31	Unlock event on LAP-S12 under the student's account	Account SID; the laptop was in use before the change	Account session, not necessarily the student
Browser / 22:38	Portal login page visit and abstract form cache	Profile and URL; preparation for the change	Cache may pre-date the incident
E-mail / 22:42	Provider notice: “new sign-in to your portal account”	Message-ID and headers; candidate authentication event	Verify headers against provider logs
Mobile / 22:44	Authenticator approval entry	MFA transaction ID; approval linked to the login	Approval context still ambiguous
Cloud portal / 22:48	Update call with submission ID PB-HK-5521 and file hash	Server-side proof of what was accepted	Shows the account, not the human
Network / 22:47	DHCP lease for the campus gateway IP	IP and time mapping to a public address	NAT may represent many users
Chat / 22:55	Message to a teammate: “abstract updated”	Ties content to a conversation	Account could be impersonated

The strongest comparisons use stable values such as the portal submission ID, uploaded-file hash, device identifier or verified account SID. Similar filenames, IP addresses and close timestamps are supporting clues only. I would test three explanations: the student made the change; someone else used stolen credentials or an existing session; or a shared device or authorised workflow caused the activity.

Q37. Major challenges when evidence is spread across devices and cloud services

1. **Different permissions:** cloud providers and personal devices may each need separate legal or administrative authority.
2. **Short-lived evidence:** provider logs, browser sessions, app notifications and chats may be deleted or rotated.
3. **Encryption:** modern phone and computer security features can limit what a tool can extract.
4. **Uncertain identity:** an account name, IP address or MFA approval does not always identify the human using the service.
5. **Different clocks:** providers and devices record time in different zones, sometimes with delays or wrong settings.
6. **Changing data formats:** browser, messaging and cloud export formats change, and no tool parses all of them.
7. **Too much unrelated data:** investigators must avoid reading private messages and files that are not part of the incident.
8. **Chain of custody:** screenshots or exports without a collection record are hard to verify later.
9. **Duplicate traces:** one cloud event can synchronise to several devices and look like several activities.

Q38. Final investigative conclusion with evidential tiers

Tier	Possible conclusion in scenario C5	What remains necessary
Direct platform evidence	A validated server audit would show the account accepted submission PB-HK-5521 at 22:48	Original provider log, object record and query provenance
Corroborating evidence	Browser cache, the sign-in e-mail and the mobile approval may align with the portal event	Device and account ownership, authentic headers, time normalisation
Unresolved questions	Which person controlled the session, how the login was obtained and whether the approval was voluntary	Independent interviews, safe credential review and any lawful additional records

My provisional conclusion would be limited to what the evidence can support. If the portal's own logs confirm the change, I could say that the student's account was used to alter the registration. I could not say that the student personally made the change without further corroboration. The report should explain every reasonable alternative and recommend proportionate steps, such as preserving logs and then reviewing session and MFA settings.

When the laptop, browser, messages, phone and portal agree on the sequence, confidence in the timeline rises. That agreement still has to be kept separate from the harder question of who acted and why.

REFERENCES / APPLICABLE BACKGROUND GUIDANCE

Reference Notes and Submission Checklist

The assignment handout is the main source for the questions and deliverables. The following references give background on digital-evidence handling, imaging and shadow copies. They support the proposed workflows and do not replace approved laboratory procedures.

- [1] NIST SP 800-86, Guide to Integrating Forensic Techniques into Incident Response (2006). https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-86.pdf
- [2] NISTIR 8387, Digital Evidence Preservation: Considerations for Evidence Handlers (2022). https://doi.org/10.6028/NIST.IR.8387
- [3] National Institute of Justice, Test Results for Digital Data Acquisition Tool: IXimager Version 2.0 (2007). https://nij.ojp.gov/library/publications/test-results-digital-data-acquisition-tool-iximager-version-20-feb-01-2006
- [4] Microsoft Learn, Volume Shadow Copy Service (VSS). https://learn.microsoft.com/en-us/windows-server/storage/file-server/volume-shadow-copy-service
- [5] Forensics Wiki, IXimager and IXimager file formats. https://forensics.wiki/iximager/ and https://forensics.wiki/iximager_file_formats/
- [6] NIST SP 800-101 Rev. 1, Guidelines on Mobile Device Forensics (2014), as background for Q34.

Final review before submission

- All five case studies and all 38 questions are answered in the order of the handout.
- Each assignment has a distinct, clearly titled scenario with its assumptions stated.
- The required flowcharts, evidence tables, timelines, report outline and sample chain-of-custody form are included.
- Sample evidence, times and custody entries are identified as examples, not actual tool output.
- The answers explain forensic integrity, attribution limits, alternative hypotheses and responsible evidence handling.
- The cover page has the student name, register number and submission date; check them before handing in.

As stated in the handout, the submission deadline is 16 October 2026, before 4:00 p.m.